

XGBoost Churn Prediction Model

Model card & methodology · 93,358 customers · Olist Brazilian e-commerce · label: no purchase within 180 days

0.868	92.5%	84.4%	0.883
ROC AUC (hold-out)	Recall @ threshold 0.30	Precision	F1

A churn model is only worth building if someone acts on it. This one is delivered as a ranked watchlist inside a Power BI dashboard, with the driver breakdown attached to every flagged customer, so a retention team can start calling on Monday morning.

1 · Problem framing

The platform keeps 25.9 % of its 93,358 customers. Retention spend has to be aimed, not sprayed, so the model answers a single operational question: **which customers should be contacted first?** That framing decides the metric policy — in retention, a missed churner costs far more than a wasted outreach, so the operating threshold is tuned for recall rather than accuracy.

2 · Data and feature engineering

38 features per customer, assembled from three sources and collapsed to one row per customer_unique_id:

Feature group	Features	Meaning
Spend	total_spent · order_count	historical value and depth of relationship
Satisfaction	avg_review · high_satisfaction	mean review score · share of ≥4-point reviews
Delivery quality	late_rate · avg_delay_days	share of late orders · mean delay in days
Payment behaviour	avg_installments · credit_card_rate	instalment usage · card share
Seller & freight	n_sellers · avg_freight	number of distinct sellers · mean freight cost
Breadth	unique_categories	distinct product categories purchased
Geography	st_* (27 one-hot)	customer state

3 · The label-leakage diagnosis (the most useful finding)

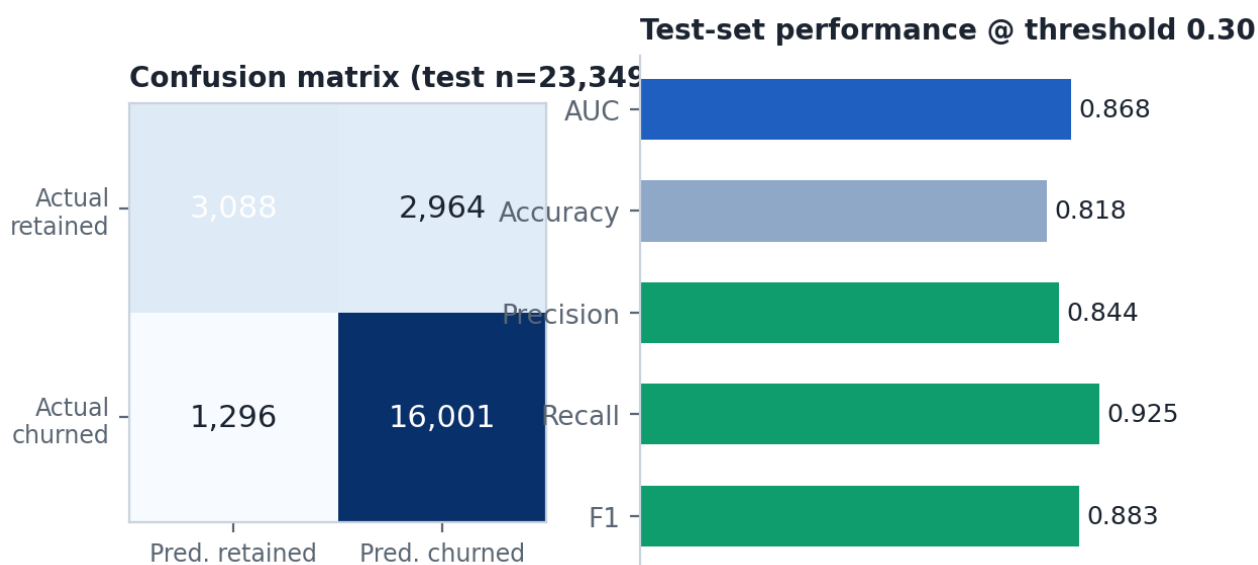
The first model scored AUC 1.000. That is not a good model, it is a tautology: the label is defined as `days_since_last > 180`, and the feature set contained `days_since_last`. Feeding the definition of the label to the learner guarantees a perfect score and zero business value. Removing recency dropped AUC to 0.677 — that is the honest, behavioural predictive power — and adding payment, seller and freight features lifted it to 0.868.

Version	Feature set	AUC	Reading
v0 naive	includes days_since_last	1.000	tautological — the answer was in the features
v1 leakage-free	behavioural features only	0.677	true predictive power
v2 enriched	+ payment / seller / freight	0.868	shipped model

Detecting and reporting the leak is itself the deliverable: a defensible 0.868 is worth more than an impressive 1.000 that collapses in production. A time-separated variant (predict 2018 activity from 2017 behaviour) also confirmed that only 1.5 % of observed customers repurchase within the window — an Olist business fact, not a modelling failure.

4 · Training and validation

Setting	Value	Rationale
Algorithm	XGBoost classifier, binary:logistic	handles mixed scales and interactions
max_depth / learning_rate	7 / 0.03	medium complexity, small steps
n_estimators	800	paired with the low learning rate
subsample / colsample_bytree	0.8 / 0.8	row and column sampling against overfitting
scale_pos_weight	2.86	74 : 26 class imbalance
reg_lambda	1.0	L2 regularisation
Split	stratified 75 / 25 (70,047 train · 23,349 test)	stable class ratio
5-fold CV AUC	0.864 ± 0.002	score is not a split artefact
Threshold policy	0.30 (swept 0.30–0.70)	recall-first for retention outreach



Model evaluation — ROC, cross-validation stability and threshold sweep.

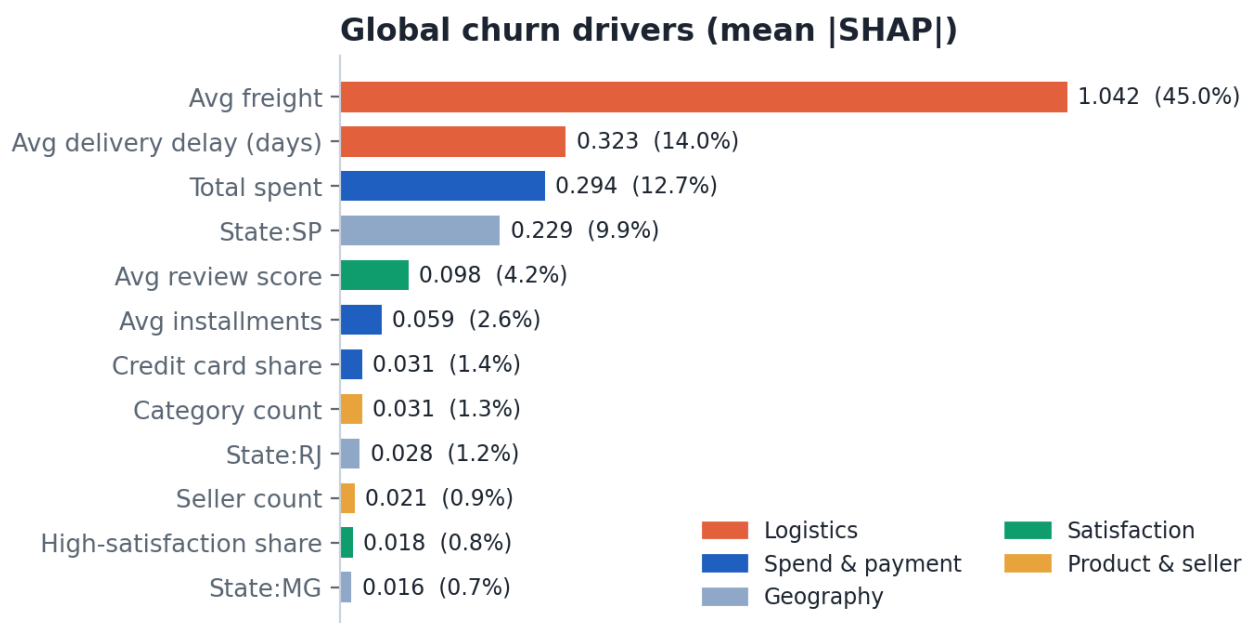
5 · Evaluation

Metric	Value	Note
ROC AUC	0.868	hold-out, leakage-free
Accuracy	0.818	reported for completeness, not the objective
Precision	0.844	at 0.30
Recall	0.925	9 of every 10 churners are flagged
F1	0.883	balance of the two

6 · Explainability — why each customer is flagged

Global SHAP ranking, and a per-customer TOP-1 / TOP-2 / TOP-3 driver breakdown carried into the dashboard watchlist:

Rank	Feature	mean SHAP	Business reading
1	avg_freight	1.042	freight burden — cost friction kills the second purchase
2	avg_delay_days	0.323	delivery reliability — matches the causal finding
3	total_spent	0.294	value concentration
4	st_SP	0.229	São Paulo concentration effect
5	avg_review	0.098	satisfaction
6	avg_installments	0.059	financing behaviour
7	credit_card_rate	0.031	payment mix
8	unique_categories	0.031	breadth
9	st_RJ	0.028	Rio de Janeiro effect
10	n_sellers	0.021	marketplace fragmentation



SHAP driver ranking — delivery quality and freight dominate, which is what the causal analysis independently concluded.

7 · Deployment and monitoring

- **Scoring** — batch score on every refresh; each customer carries a churn probability, a risk tier and its TOP-3 drivers.
- **Consumption** — a Power BI watchlist sorted by probability, with the intervention cost and expected value beside each name.
- **Monitoring** — AUC / recall are watched on a rolling hold-out; a drift in the freight or delay distributions is the early warning, since those two features carry most of the signal.
- **Retraining trigger** — recalibrate when recall at the operating threshold falls below 0.85, or when the positive base rate moves more than 5 pp.
- **Caliber lock** — the label definition is frozen (dormancy > 180 days, sensitivity-checked at 30/60/90/120/180/270/365) so model drift is never confused with a caliber change.

8 · Limitations, stated plainly

- The label is behavioural dormancy, not a stated cancellation — a customer who left for reasons we cannot observe still counts as churned.
- 29 % of buyers flagged as “repeat” are single-session split orders; the honest cross-period repeat rate is 2.10 %, so classes built on repeat behaviour are thinner than they look.
- Feature importances are predictive, not causal. The causal claims live in the evidence brief, where the delivery effect is identified with PSM / IV / DML.
- Geographic one-hot encoding ties the model to the Brazilian state list; a new market means a retrain, not a patch.

Outcome: recall 92.5 % at a policy threshold of 0.30 means nine in ten future churners are surfaced before they go quiet — and every one of them arrives with a reason attached.

How this was produced. Human-led method, AI-assisted execution. The business problem, the analysis framework, the choice of method and the validation of every result are mine; AI handled implementation and drafting. Each figure is reconciled by an automated gate before it ships, and no claim reaches a client document without a human having checked it against its source snapshot.